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Controlling stormwater detention ponds under partial observability

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ABSTRACT

Stormwater detention ponds play an important role in urban water management for collecting and conveying rainfall runoff from urban catchment areas to nearby streams. Their purpose is not only to avoid flooding but also to reduce stream erosion and degradation caused by the direct discharge of pollutants to the stream. We model the problem of controlling the discharge rate of water from the ponds as a partially observable hybrid Markov decision process and subsequently use Uppaal Stratego for synthesizing safe and near optimal control strategies. The generated strategies are based on noisy sensor measurements of the water height in the pond, hence the underlying system is only partially observable. We present results analyzing how sensitive the synthesized strategies are with respect to the accuracy of the measurement sensors in both offline and online settings. These types of analyses not only provide insight into the robustness of the generated strategies, but they can also be used for deciding on which measurement sensors to use, thereby balancing sensor cost and accuracy.

1. Introduction

In urban water management, stormwater detention ponds play a crucial role by collecting and directing rainfall runoff from catchment areas to nearby streams [1]. Their objective extends beyond preventing flooding; it also includes mitigating stream erosion and degradation resulting from the direct discharge of pollutants into the stream [2].

Standard solutions for managing stormwater detention ponds employ static controllers to regulate the discharge of water [3]. These types of controllers fail to exploit the full potential of the water system as demonstrated in [4], where formal methods are used to synthesize safe and optimal control strategies using Uppaal Stratego [5]. For synthesizing control strategies, [4] models the water system as a hybrid Markov decision process (HMDP), based on which reinforcement learning is used to find a strategy that minimizes a cost function balancing the consequences of discharging pollutants to the stream (optimality) with the consequences of pond overflows (safety).

The control strategies synthesized in [4] rely on full observability of the system, including the actual water height in the pond. In real-world settings, water measurements are based on noisy and potentially faulty sensors leading to incomplete and partial observability. In this paper, we extend the results in [4] by modeling the stormwater detention pond, covering also catchment areas and weather forecasts, as a partially observable hybrid Markov decision process, where sensors and sensor noise are explicitly represented. We compare the synthesized strategies with those produced by [4] and we present analyses of how sensitive the

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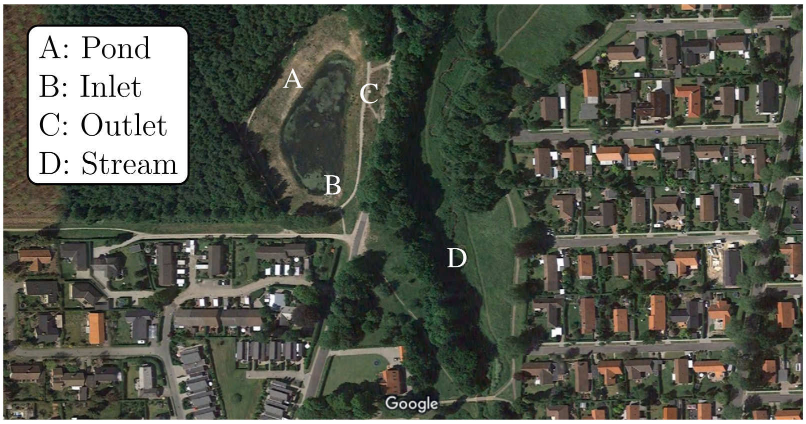

Fig. 1. A stormwater detention pond in Vilhelmsborg skov near Aarhus, Denmark.

synthesized strategies are with respect to different types of sensor noise. These analyses not only provide insight into the robustness of the control strategies, but can also be used for designing new physical sensor systems, where one typically needs to find an appropriate balance between the cost and accuracy of the sensors.

This paper is an extended version of the conference paper [6] with the following additions. First, we include a formal definition of the mathematical modeling framework HMDP, accompanied by a full model specification of the stormwater detention pond as well as its relevant subsystems, such as the surrounding urban catchment area and the weather forecast. Second, we conduct analyses and experiments on generating control strategies for the detention pond under conditions of partial observability, utilizing only water level measurements through noisy sensors. Third, by utilizing the STOMPC framework [7], where UPPAAL STRATEGO and EPA-SWMM [8] (a domain-specific simulator) are combined, we expand our analysis to include an online Model Predictive Control (MPC) setting.

This paper is structured as follows. In Section 2, we give an introduction to stormwater detention ponds and urban water management, and in Section 3 we introduce the mathematical framework used for modeling such systems. In Section 4 we describe the control problem and the system model, which form the basis for the offline and online experimental results discussed and analyzed in Section 5. Finally, in Section 6, we conclude the paper with discussions and further research challenges.

2. Stormwater detention ponds

In urban water management, a stormwater detention pond is a specially designed basin or pond for collecting rainwater from urban areas, such as streets and parking lots, before discharging the water into streams (see Fig. 1). Stormwater detention ponds primarily aim to regulate and oversee the discharge of stormwater runoff into downstream water bodies. By temporarily retaining the runoff of urban catchment, detention ponds aid in the elimination of pollutants from the stormwater runoff, leading to an improvement in water quality prior to its discharge into the nearby stream. Moreover, the detention ponds facilitate a gradual drainage process, which helps diminish the runoff's peak flow rate and prevent downstream erosion and flooding.

In the event of heavy rainfall, if the volume of water entering the detention pond from the urban area exceeds its holding capacity, it will cause the pond to overflow. This can subsequently result in flooding downstream. By monitoring the water level in the pond, sensors can help control the water flow and prevent flooding in advance. For this reason, industries install sensors at the site to monitor and control their manufacturing processes. These sensors can measure various parameters such as temperature, pressure, pH, and water level [9].

Sensor imperfection encompasses any deviation from optimal sensor performance, which may include measurement errors or inaccuracies. Noise is one of the most common types of sensor imperfection, resulting from random signal fluctuations caused by sources like thermal changes or electrical interference. These fluctuations may challenge obtaining precise signal measurements [10, 11]. Sensor imperfections can impact control strategies because precise sensor measurements are essential for determining a system's condition and the most appropriate control decisions in a given situation. As a result, sensor imperfections can lead to suboptimal or hazardous control decisions, which can have significant consequences in safety-critical fields.

An example of a stormwater detention pond with a sensor installed is schematically illustrated in Fig. 2. Rainwater is collected from an urban area, which is a geographic region characterized by a high population density and human-made infrastructures such as buildings, roads, transportation networks, and other urban amenities. The rain subsequently enters the pond through inlet Q_{in} before being diverted to a nearby stream through an orifice that can be used to control the discharge rate Q_{out}. The water level in the pond is denoted by w, and if it reaches the maximum water level, denoted by W, it will trigger an emergency overflow from the pond. The pond is equipped with a sensor providing noisy measurements w_{obs} of the water level, which can be used to control the orifice size.

When defining or synthesizing a control strategy for the orifice, the main objective is to prevent overflow, while at the same time allowing for sedimentation of pollutant particles from the urban runoff before discharging the water into the stream.

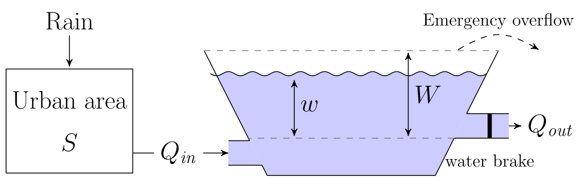

Fig. 2. Schematic illustration of a water pond. The actual water height is measured by a (noisy) sensor. The original illustration is in [4].

3. Preliminaries

3.1. Hybrid Markov decision process

We model a control problem of stormwater detention pond using a hybrid Markov decision process (HMDP) in which the (uncontrollable) state of the environment changes probabilistically [12,13].

Definition 1. A hybrid Markov decision process $\mathcal{M}$ is a tuple (C, U, X, F, δ), where:

1. The controller C is a finite set of (controllable) modes $C = \{c_1, \ldots, c_k\}$.
2. The uncontrollable environment U is a finite set of (uncontrollable) modes $U = \{u_1, \ldots, u_l\}$.
3. X is a finite set of continuous (real-valued) variables $X = \{x_1, \ldots, x_n\}$.
4. $\forall m \in C \times U$, function $F_m : \mathbb{R}_{>0} \times \mathbb{R}^X \to \mathbb{R}^X$ describes the flow-function, which is the evolution of the continuous variables over time in the mode m.
5. δ is a family of probability functions $\delta_\gamma : \mathbb{R}_{\geq 0} \times U \to [0, 1]$, where $\gamma = (c, u, \boldsymbol{x}) \in (C \times U \times \mathbb{R}^X) = \mathbb{C}$ is a configuration and vector $\boldsymbol{x}$ is a state of the continuous variables X. More precisely, $\delta_\gamma(\tau, u')$ is the probability that in the global configuration $\gamma = (c, u, \boldsymbol{x})$, u will change to u' after a time delay τ.

As detailed in Section 4, for our particular domain the orifice setting of the stormwater detention pond is represented by the controller C, which can be changed periodically. The duration and the intensity of the rain events will be modeled by the uncontrollable environment variable U, and the flow function F will model the evolution of the water in the pond as captured by the real-valued state variables X. The Markov property is implicitly included in Definition 1, where the progression of the system's state in the composite mode is solely influenced by the present global configuration $\gamma = (c, u, \boldsymbol{x})$ and is independent of the preceding sequence of events. The stochastic nature of the weather is represented by the probability function δ and, in the rest of the paper, the controller is only allowed to change mode at a fixed frequency (corresponding to period $P \in \mathbb{R}_{\geq 0}$), say, every one hour. The general definition of an HMDP allows for the modeling of time as a state variable, and time can be included among the continuous (real-valued) variables X presented in Definition 1.3.

A *run* for an HMDP $\mathcal{M}$ is a sequence π of configurations and relative time-delays:

$$\pi = \gamma_0 :: \tau_1 :: \gamma_1 :: \tau_2 :: \gamma_2 :: \cdots, \tag{1}$$

where $\gamma_i = (c_i, u_i, \boldsymbol{x}_i) \in \mathbb{C}$, $\tau_i \in \mathbb{R}$, and for all i either

- the environment u_i changes to a new state u_{i+1} in γ_{i+1}, where $c_{i+1} = c_i$ and $\delta_{\gamma_i}(\tau_{i+1}, u_{i+1}) > 0$ or
- the controller changes to a new mode c_{i+1} in γ_{i+1} at the end of a period, i.e., $\sum_{j \leq i+1} \tau_j = n \cdot P$, for some n, and $u_i = u_{i+1}$.
- τ_i can be any real value and it does not need to be discretized in any manner.

3.2. HMDP strategies

A *strategy* for an HMDP $\mathcal{M}$ is a function $\sigma : \mathbb{C} \to 2^C$ that given the current configuration $\gamma = (c, u, \boldsymbol{x})$ returns a set of controllable modes. A strategy is said to be *deterministic* if only one control mode is returned for each configuration $\gamma \in \mathbb{C}$. Otherwise, the strategy is *non-deterministic*.

For a given strategy σ, the HMDP $\mathcal{M}$ defines a stochastic process denoted $\mathcal{M} \restriction \sigma$. A run of an HMDP $\mathcal{M}$ under strategy σ is run where the control modes change at a fixed frequency (corresponding to period $P \in \mathbb{R}_{\geq 0}$) according to σ. Formally, if the controller switches to a new mode c_{i+1} at the end of a period, it must hold that $c_{i+1} \in \sigma(c_i, u_i, F_{(c_i,u_i)}(\tau_{i+1}, \boldsymbol{x}_i))$.

For an objective function $f : \mathbb{C} \to \mathbb{R}$, we evaluate the quality of a strategy σ in terms of the expected value of f over the space of runs of $\mathcal{M} \restriction \sigma$ starting in configuration γ_0 and with time horizon $H \in \mathbb{R}_{\geq 0}$: $\mathbb{E}^{\mathcal{M},\gamma}_{\sigma,H}(f)$. In this paper, we define f to be an accumulative reward (cost) function, i.e., $f = \int_0^H r(t)\mathrm{d}t$, where $r(t)$ is the reward at time t, so $r(t)$ needs to be integrable over the interval $[0, H]$. An *optimal strategy* σ_{opt} is a strategy that maximizes (or minimizes) $\mathbb{E}^{\mathcal{M},\gamma}_{\sigma,H}(f)$. For the control problem modeled of our domain, we will define an objective function balancing sedimentation and overflow as captured by configurations $\mathbb{C}$ in Section 4.6.

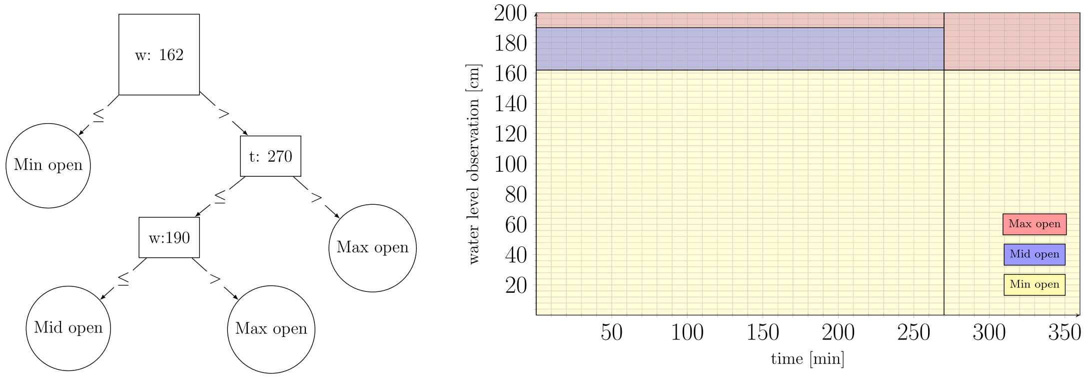

Fig. 3. Schematic illustrations of a simplified decision tree for representing control strategies.

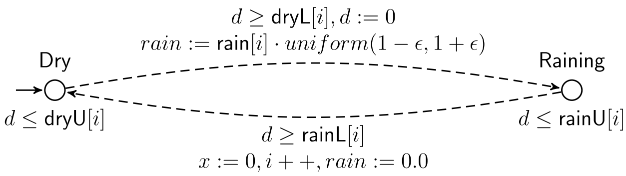

Fig. 4. Model of the weather forecast.

3.3. Uppaal Stratego

For synthesizing a strategy for controlling the discharge amount of water (Q_{out}) from the pond, we rely on the tool Uppaal Stratego [5]. Uppaal Stratego provides methods for synthesizing safe and near-optimal strategies for hybrid Markov Decision Processes (MDP), including both timed automata and continuous-space MDPs. For synthesizing a strategy, Uppaal Stratego relies on reinforcement learning techniques optimizing a specific optimization objective. For synthesizing control strategies for continuous-space MDPs, the tool uses the methods from [14,15] to partition a continuous state-space into a discrete one.

Using a tool based on formal methods provides us the advantage of having a modeling language with formal semantics. This eases the modeling step. Furthermore, Uppaal Stratego allows for synthesizing a guaranteed safe shield used for reinforcement learning [16] as well as statistical model checking of the controlled HMDP.

The strategies obtained by Uppaal Stratego, as documented in [12], can be visualized as decision trees, as exemplified in the left-hand side of Fig. 3. When $w \leq 162$, the optimal control decision is to open the controller to the minimum size. If $w > 162$, the control decision depends on the variable t. If $t \geq 270$, the optimal control decision is to set the controller to maximum size. Conversely, if $w \leq 190$, the optimal control decision is to set the controller to the medium size. If $w > 190$, the optimal control decision is to set the controller to the maximum size. The right-hand side of Fig. 3 shows the state-space partitioning of the decision tree with decision variables t and w, which are observed by the system. Using the parameter t on the x-axis and w on the y-axis, one can quickly gain an overview of the decisions without going through the decision tree.

4. Stormwater detention ponds: system model

The main (safety) objective of the pond controller is to prevent the pond from overflowing, which can have adverse affects on both the surrounding environment and infrastructure. The secondary (optimization) objective is to sediment and filter out urban particles collected in the pond in order to prevent contamination of the water environment in the stream. The longer the water stays in the pond before being discharged, the higher the degree of sedimentation of water pollutants, resulting in a reduction of discharged pollutants.

The water level in a stormwater detention pond, which is our system's control target, is primarily influenced by the amount of rainwater that falls in the surrounding urban area and flows across the ground surface, known as the urban catchment. We therefore require a rain and urban catchment model to determine the inflow to the pond. The pond's controller manages the outflow by adjusting the orifice size that carries water out from the pond to the nearby stream. The system model therefore consists of multiple subsystem components, which are described in the following sections.

4.1. Rain fall model

Fig. 4 illustrates the rain model proposed in [4]. The model generates an uncontrollable input rain to the system that can also be seen as a weather forecast used during the learning process; the model 1) provides bounds on the duration of dry and rainy periods, and 2) incorporates uncertainty in the intensity of the rainfall. To represent uncontrollable transitions, where uncontrollable variables

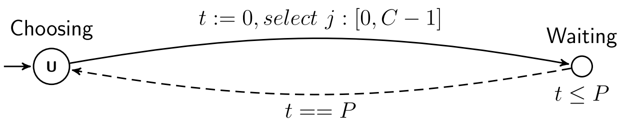

Fig. 5. Model of the orifice controller for the pond, which can change modes periodically, regulating the water discharge from the pond.

$u \in U$ are selected, we use dotted lines, e.g., for indicating the transition between Dry and Raining. The clock d tracks the current time duration and resets when the state changes. The duration of the dry and raining intervals are chosen uniformly at random within the specified bounds, dryL, dryU, rainL and rainU. When transitioning from the Dry to the Raining state during a specific period represented by i, the rain intensity, denoted as $rain \in \mathbb{R}$, is selected uniformly at random within the range of $rain[i] \cdot (1-\epsilon)$ to $rain[i] \cdot (1+\epsilon)$, where $\epsilon = 0.1$ serves as the uncertainty factor. The rain intensity is assumed to remain constant throughout the period. For the experiments, the duration and intensity of the rain events are based on historical rain data obtained from the Danish Meteorological Institute (DMI).

4.2. Urban catchment model

We use the one-layer linear reservoir model (the surface storage of the simplified 'Nedbør Afstrømnings Model' (NAM)) [17] to describe the dynamics of the water in the urban catchment area. In our model, we assume a uniform distribution of water across the entire urban catchment area. We account for the dynamics of water volume S over time t using Equation (2), where k denotes the surface reaction factor.

$$\frac{\mathrm{d}S}{\mathrm{d}t} = rain - kS. \tag{2}$$

From Equation (2) we see that changes in the water volume S stored in the urban catchment area depends on the difference between the amount of rainfall in the area and the water dissipating from it (determined by the water volume and the surface reaction factor k).

4.3. Model of the stormwater detention pond

The differential equation in Equation (3) characterizes the change in water volume V over time t in the pond. This volume change is determined by the difference in the inflow Q_{in} and outflow Q_{out} of the pond. In our system model, we assume that only the urban catchment area provides inflow into the pond, denoted as Q_{in}, and disregard other potentially contributing sources such as rainwater and melted snow that might enter directly into the pond [18]. The outflow Q_{out} of the pond changes according to the dimensions of the opening and the gravitational pull exerted by the water level within the pond.

$$\frac{\mathrm{d}V}{\mathrm{d}t} = \begin{cases} 0 & \text{if } (w = 0) \wedge Q_{out} \geq Q_{in}, \\ 0 & \text{if } (w = W) \wedge Q_{in} \geq Q_{out}, \\ Q_{in} - Q_{out} & \text{else} \end{cases} \tag{3}$$

The first boundary case in Equation (3) occurs when the outflow is greater than the inflow, and the water level is zero (representing an empty pond). The second case describes an overflow situation during which the inflow is greater than the outflow, and the water level is at the maximum level. Hence no additional water can accumulate in the pond.

4.4. Controller model

Fig. 5 shows the model of the controller. The pond's opening size j can be changed periodically to control the outflow of the pond. The controller monitors the current duration of the control period using a locally operating clock t, which resets periodically to 0. While in the Waiting location, the controller waits for P time units before repeating this process on an hourly basis. Due to physical constraints of the available outflow control devices for stormwater ponds, the controller is only able to choose from a set of C control modes (where C is typically 1, for static control, or a low number). Location Choosing is labeled as an urgent location (indicated with U) in which time cannot progress. Hence, a controller is forced to make a control decision periodically. To represent controllable transitions, where controllable modes $c \in C$ are selected, we use solid lines, e.g., for the transition from Choosing to Waiting.

4.5. Sensor model

The water level in the pond is measured at every period P using a sensor that provides a noisy signal of the actual water level in the pond. Specifically, we assume an additive noise model [9]:

$$w_{obs}(t) = w(t) + noise(t), \tag{4}$$

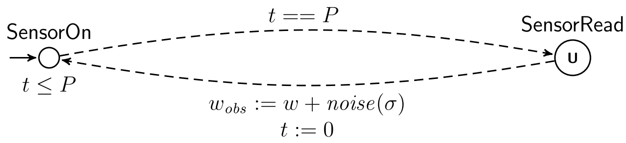

Fig. 6. Model of a sensor providing noisy measurements of the water level. The sensor measurements are updated every period P.

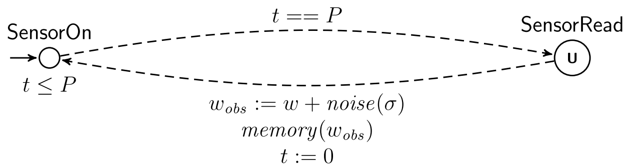

Fig. 7. Model of noisy sensor that retains the most recently recorded measurements in memory. Similar to the standard sensor (see Fig. 6), the sensor measurements are updated every period p.

where the noise is Gaussian distributed with mean 0 and standard deviation σ, i.e., $noise \sim \mathcal{N}(0, \sigma^2)$. To avoid confusion in the remainder of the paper, σ always refers to the standard deviation of the noise, while a strategy is always accompanied with a subscript, like σ_{opt}. Fig. 6 shows the sensor model capturing the relationship between the actual water level w and the observed water level w_{obs}. The constraint on the arrow from SensorOn to SensorRead indicates that this transition occurs periodically when t reaches P. On the other hand, the arrow from SensorRead to SensorOn represents the observed value w_{obs} as defined by the additive noise model above.

4.5.1. Memory equipped sensors

Sensors are becoming more advanced and there exist several types of sensors that possess memory or data storage capabilities [19]. One example is data logger sensors, which are capable of recording and storing information over a specific period. These types of sensors are frequently employed in environmental monitoring applications to keep track of factors such as humidity, pressure, and water levels. In addition to the utilization of memory in conjunction with sensors, there have been recent efforts to optimize sensor memory management, as evidenced by studies such as [20], which explore the use of lossless memory to improve efficiency.

The past recordings stored in such sensors may provide more fine-grained insight into the true state of the system being modeled, hence also proving an opportunity for improved control synthesis. We extend the sensor model introduced above with a memory component operating as a fixed-size FIFO queue (implemented with the function $memory(w_{obs})$ in Fig. 7). During synthesis, the controller has access to all N stored values.

4.6. Objective function

The objective of the control problem is to prevent overflow from the stormwater detention pond, while simultaneously ensuring that pollutants from the urban runoff area are maximally sedimented instead of being discharged into the stream.

We model these two conflicting objectives using a cost function c defined by the differential equation:

$$\frac{\mathrm{d}c}{\mathrm{d}t} = \underbrace{\alpha \times \left(\frac{w}{W}\right)^{\beta}}_{\text{Part 1: overflow}} + \underbrace{\left(1 - \frac{w}{W}\right)}_{\text{Part 2: sedimentation}} \quad . \tag{5}$$

Equation (5) represents the cost function c, which incorporates components to penalize both pond overflow (Part 1) and water accumulation in the pond based on sedimentation levels (Part 2). Specifically, in Part 1 of Equation (5), we gradually apply a stronger penalty as the water level approaches W by exponent β, rather than applying a fixed penalty when surpassing W. Therefore, when the water level w is significantly below W, Part 2 (Sedimentation) is dominating, but as the water level gets closer to W, the penalization for overflow (Part 1) is progressively strengthened. The balance between Part 1 and 2 can be tuned by coefficient α.

The formulation of the control problem can now be defined as synthesizing an optimal strategy σ_{opt} that minimizes the expected cost:

$$\sigma_{opt} = \arg\min \mathbb{E}^{\mathcal{M},\gamma}_{\sigma_{obs},H}(c). \tag{6}$$

The tool UPPAAL STRATEGO is used for strategy synthesis, relying on sampling-based approximations for estimating the expectations together with reinforcement learning-based methods [12].

5. Experimental results: sensor-based control synthesis

For the model presented in Section 4, the overall goal is to synthesize a control strategy that minimizes the expected cost as defined by the cost function in Equation (5). Specifically, we analyze the effect of synthesizing controllers under partial observability of the system, while also considering the effect of

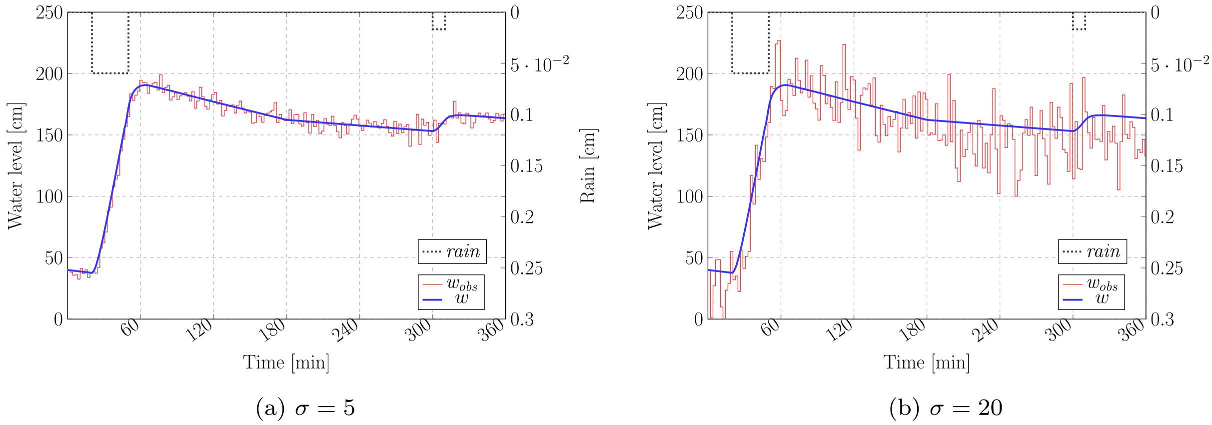

Fig. 8. Comparison of the observed water level (w_{obs}) and the real water level (w) according to the levels of noise.

Table 1
The weather forecast used for learning in the experiments.

State	Lower bound [min]	Upper bound [min]	Rain intensity [cm/min]
Dry	15	25	0
Raining	25	35	$0.05973 \pm \epsilon$
Dry	230	270	0
Raining	5	15	$0.01578 \pm \epsilon$
Dry	∞	∞	0

- varying sensor quality as encoded by the sensor noise,
- having more elaborate sensors with memory banks containing previous water measurements, and
- having multiple low-quality controllers as replacement for a single high quality controller.

The initial analyses are conducted in an off-line setting that is afterward extended to an online model predictive control scenario. The experiments are reproducible using the code package available online [21]. This package comprises both the model and accompanying scripts.

5.1. Experimental setup

The size of the orifice of the pond has three different discrete settings that can be set by the controller: 1/7, 4/7, and 7/7 of the fully opened size. For performing off-line control synthesis we consider a 6-hour prediction period, where the control setting for the orifice can be changed every hour. The strategies are synthesized under partial observability of the system: the only information available is the sensor readings and the system time. In our experiment, sensor noise is represented by σ, and as this value increases, the noise affecting the observed water level (w_{obs}) also increases. As can be seen in Fig. 8, $\sigma = 5$ and $\sigma = 20$ applied w_{obs} are presented, and it is observable that the noise increases with the increasing value of σ.

We utilize a weather forecast incorporating two major rainfall events for the synthesis of learning controllers, as detailed in Table 1. This weather forecast is generated using the system model (Fig. 4) and its values are derived from historical rain data obtained from the Danish Meteorological Institute (DMI). To model the inherent uncertainty in the weather forecast, we introduced a random variability of up to 10% in the rain intensity while the duration of the events was chosen uniformly at random based on upper and lower duration bounds. For example, the duration of the first rainfall event is randomly selected within a range of 25 to 35 minutes, featuring a rain intensity of $0.05973 \pm \epsilon$ [cm/min]. The second rainfall event's duration can range from 5 to 15 minutes, with a rain intensity of $0.01578 + \epsilon$ [cm/min].

For the controller syntheses, the objective function used for our learning is based on Equation (5) in Section 4.6. Determining appropriate values for the weights α and β depends on expert judgment and may involve some trial and error. In our study, we value safety more than sedimentation and based on preliminary experiments we set $\alpha = 1000$. Our experiments for deciding proper exponents β were conducted for exponents ranging from 10 to 100; the effect of varying β is illustrated in Fig. 9 for different values of β. Our objective was to determine the value of β at which the cost begins to increase significantly as w approaches W while at the same time the cost of overflow should remain at a low value when $w \ll W$. We selected $\beta = 70$ for our objective function because, with $\beta = 70$, the cost starts to gradually increase above 0.001 once the water level w exceeds 183 cm.

We set the learning budget in UPPAAL STRATEGO such that 140,000 total runs were generated, while for evaluation purposes, only 28,000 runs were conducted, which represent 20% of the total runs. All experiments were run on an AMD Ryzen 7 5700x with 33 GB RAM.

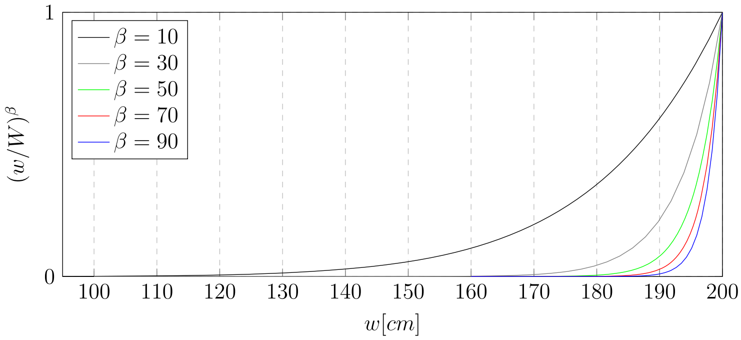

Fig. 9. The partial cost w/W with different exponents β, where W is 200 cm.

Table 2
Cost and overflow duration results based on synthesized control strategies for varying levels of sensor noise. For comparison, results are also included for a randomized controller as well as three static controllers.

	Cost [-]	Overflow [min]
No noise	731.54 ± 0.04	≈ 0
$\sigma = 5.0$	732.48 ± 0.04	≈ 0
$\sigma = 10.0$	732.55 ± 0.06	≈ 0
$\sigma = 15.0$	734.93 ± 0.25	≈ 0
$\sigma = 20.0$	737.21 ± 0.04	≈ 0
$\sigma = 100.0$	737.72 ± 0.05	≈ 0
Random	23,888.10 ± 1982.71	2.23 ± 0.55
Static-max	748.26 ± 0.04	≈ 0
Static-med	11,881.70 ± 0.23	≈ 0
Static-min	202,313.0 ± 75.93	53.10 ± 0.06

5.2. Control synthesis with different sensor types

The synthesized control strategies are based on noisy sensor readings of the water level of the pond. Below we analyze the effect of the quality of the sensor on the synthesized strategy. In particular, we consider sensors where the sensor noise can be modeled by an additive Gaussian distributed noise component with zero mean and standard deviations 5, 10, 15, and 20 (see Equation (4)). For reference, we also include results for a noiseless sensor (i.e., corresponding to the perfect observation of the water level) as well as an (almost) uninformative sensor encoded with a noise component with a standard deviation of 100. In addition, for baselines we include results for a fully randomized control strategy (denoted *random*), where the control options are selected uniformly at random as well as three static control strategies where the orifice size is fixed at the smallest (*static-min*), medium (*static-med*), or largest value (*static-max*), respectively. A fully open orifice represents a stable choice for overflow prevention, but does not serve the purpose of sedimentation. On the other hand, the static controller always selecting the least open option keeps the water level in the pond high and thus prioritizes sedimentation, but with a higher risk of overflow.

5.2.1. Quantitative results

The noise levels for the sensor were selected in order to analyze how the quality of a sensor affects the synthesized strategies as well as the results obtained from applying these strategies. Table 2 summarizes the quantitative results of the analysis in terms of overflow duration (measured in minutes) and the cost of the synthesized strategy (averaged over 10 runs), with the latter reflecting both the degree of overflow and sedimentation time (see Equation (5)). From the table, we see that the cost increases monotonically with increasing sensor noise, but that the learned strategies consistently outperforms all static controllers, irrespectively of the quality of the sensor. Furthermore, even with noisy sensors, overflow can be avoided when an optimal strategy is applied, whereas overflow occurs for ≈ 2.2 minutes when using a random controller and for ≈ 53.1 minutes when only the smallest-sized orifice is applied.

5.2.2. Qualitative results

In Fig. 10 we compare the control settings of the synthesized strategies for a fixed weather scenario but with different levels of sensor noise. The experiments were conducted ten times for the same level of sensor noise. Figs. 10a show the result of applying the synthesized strategy using a noiseless sensor. While the water level is decreasing, the synthesized controller recognizes that there is little or no risk of overflow and therefore sets the orifice size to its medium setting in order to ensure a high sedimentation rate. In comparison, the strategies synthesized using noisy sensors, as illustrated in Figs. 10b - 10f, exhibit more cautious control behaviors by generally having larger water discharge rates reflecting the amount of sensor noise.

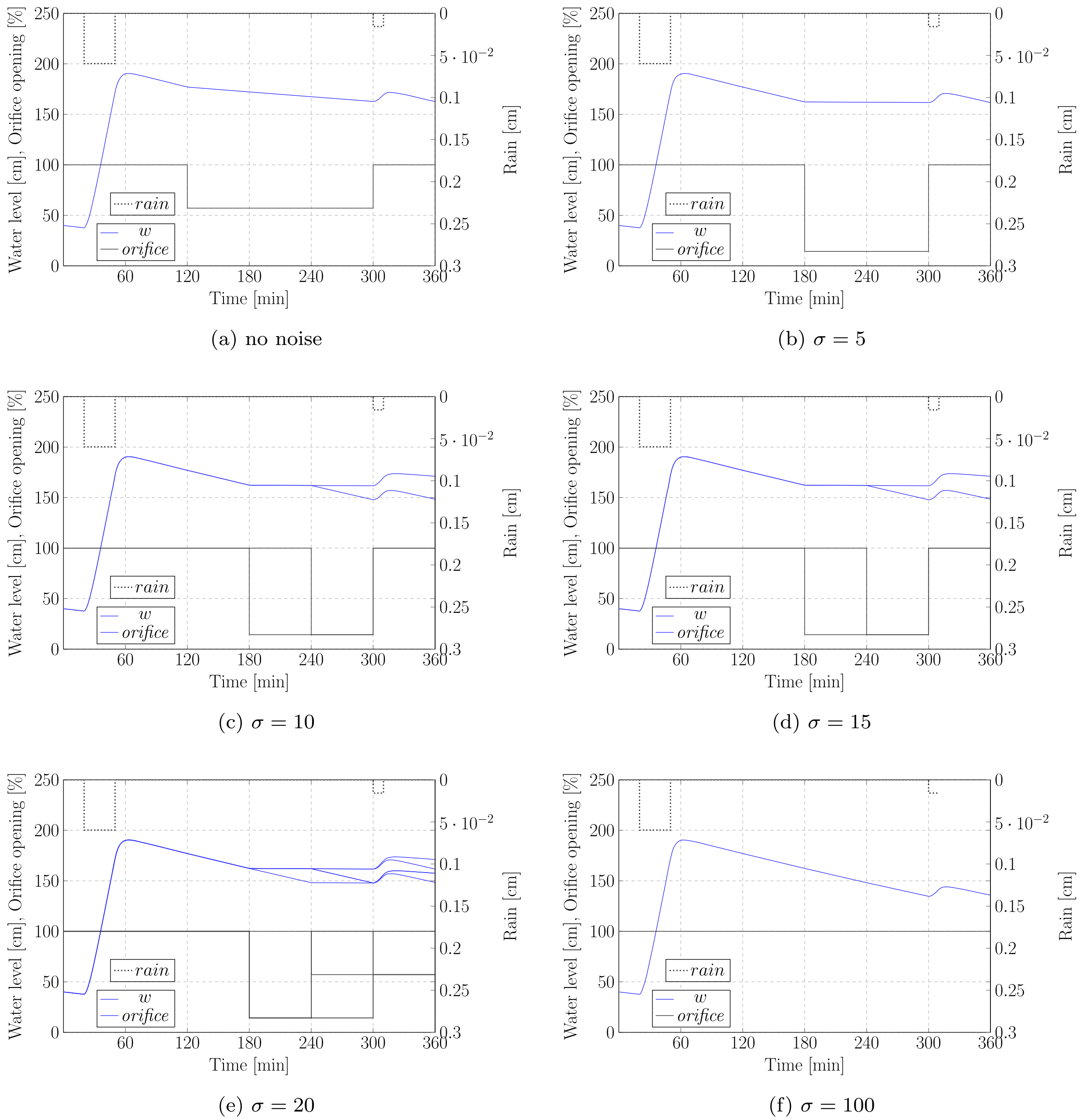

Fig. 10. Water level (w) and orifice size (opening rate (%) based on the synthesized controller for different levels of noisy sensors (each experiment is repeated 10 times).

In Figs. 10d and 10e, we see learned strategies produce different control settings for the simulations due to the uncertainty in the sensor readings. The strategy synthesized by the uninformative sensor (Fig. 10f) does not exhibit such variability over the runs. This is because the synthesized strategy recognizes that the sensor provides little information about the water level and therefore conditions the synthesized control strategy on the system time only (effectively ignoring the sensor readings). This learned strategies are illustrated in Fig. 11 as decision trees and in Fig. 12 using state space partition's; the learned strategies are based on a high-quality sensor (noise $\sigma = 5$), two medium-quality sensors (noise $\sigma = 10$ and $\sigma = 20$), and an uninformative sensor (noise $\sigma = 100$). We also see that the strategy based on the high-quality sensor exhibits some apparent inconsistencies in the control settings for the time interval $[208, 228]$. We hypothesize that this is partly a consequence of the underlying reinforcement learning algorithm not being able to sufficiently explore certain parts of the state space. Similar times of apparent inconsistencies are also seen in the other strategies being synthesized. Results of an additional offline experiment are included in Appendix A.

Table 3
The weather scenario utilized for evaluation. Unlike the weather forecast used for learning (see Table 1), the duration of Dry and Raining conditions remains constant in this evaluation and uncertainty in rainfall intensity is also not considered.

State	Duration [min]	Rain intensity [cm/min]
Dry	20	0
Raining	30	0.05973
Dry	250	0
Raining	10	0.01578
Dry	50	0

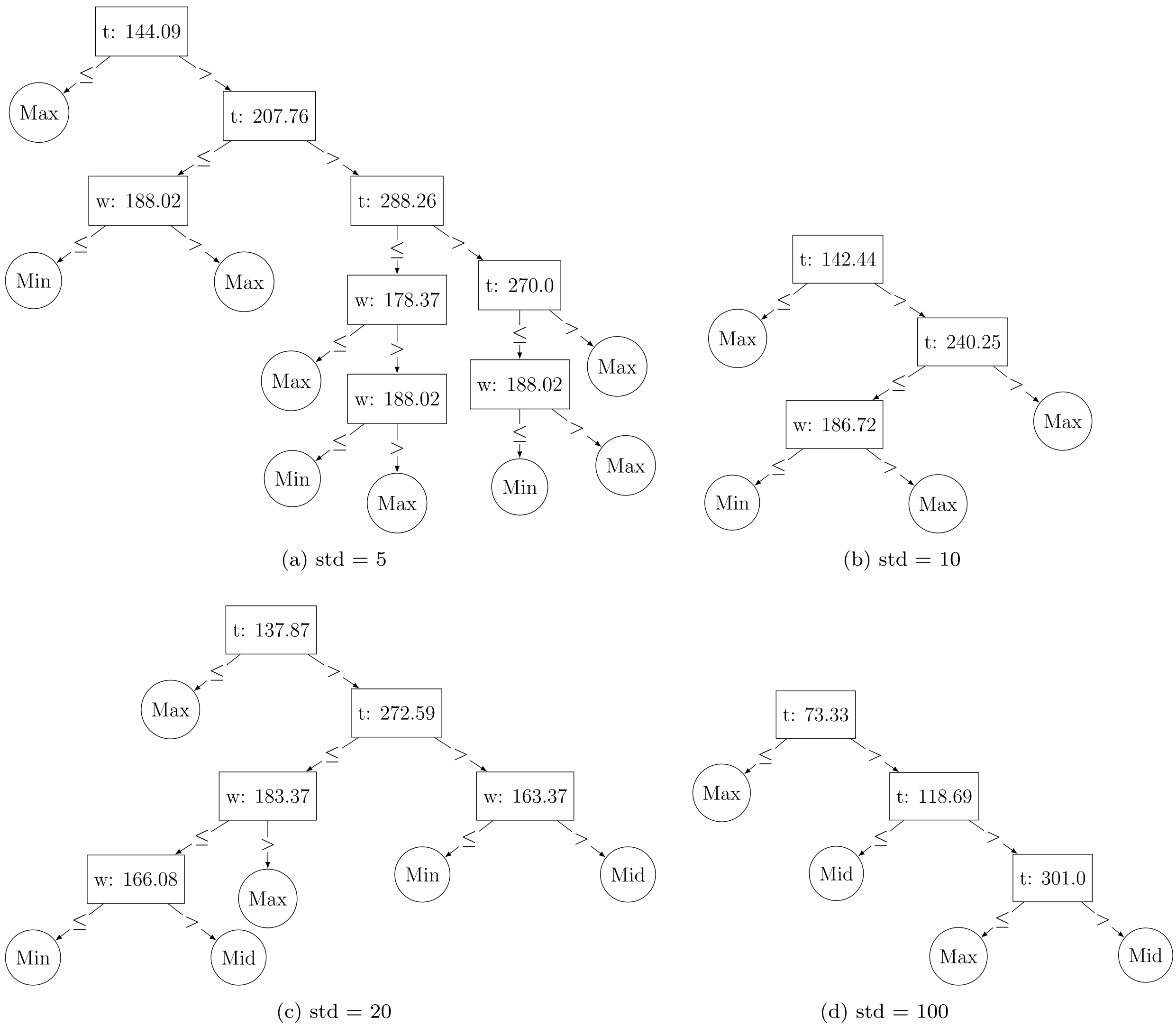

Fig. 11. Decision tree representations of the strategies synthesized based on a high-quality sensor (a: $\sigma = 5$), two medium quality sensors (b: $\sigma = 10$) and (c: $\sigma = 20$), and an uninformative sensor (d: $\sigma = 100$).

5.3. Online synthesis using model predictive control

All previous experimental results are in the setting of offline control synthesis, which uses an algorithm that takes a mathematical model of the system and a set of performance objectives as inputs to compute a control policy. The generated control policies are used

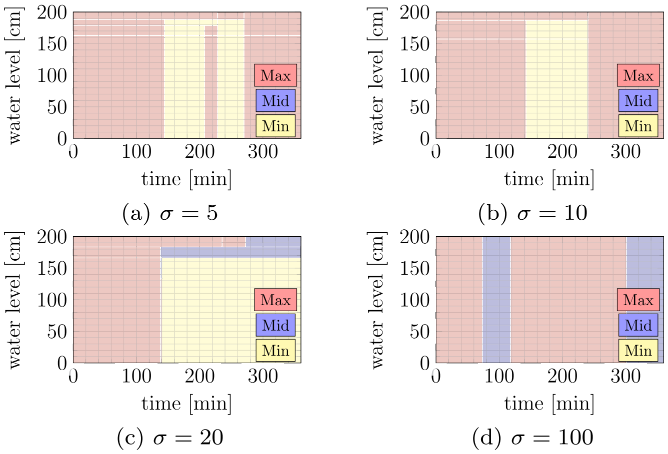

Fig. 12. State-space partitions of the strategies synthesized based on a high quality sensor (a: $\sigma = 5$), two medium quality sensors (b: $\sigma = 10$) and (c: $\sigma = 20$), and an uninformative sensor (d: $\sigma = 100$).

over the full time-horizon. If the system exhibits a cyclic behavior, one can use the synthesized controller for infinite-time horizons. This approach is often used when the system's dynamics and controller objectives are well-understood and well-defined. Examples of such applications include manufacturing, robotics, and process control [22].

Furthermore, online synthesis techniques, such as model predictive control (MPC), actively generate control policies in real-time. This is achieved by utilizing continuously updated sensor readings [23]. The distinct advantage of MPC lies in its ability to adapt: the control strategies are not static but are revised in response to changes in the system or its environment. This dynamic adaptation enables the control strategies to be effective even under new or changing conditions. Such approaches are particularly beneficial in scenarios where the system's dynamics are unpredictable or subject to variation, or when the objectives of the controller evolve over time [24]. In online synthesis, modeling inaccuracies have a diminished impact on the outcome. The reason is that the model directing the controller synthesis undergoes regular re-calibrations to accurately represent the current state of the system. This feature is particularly pertinent in the context of our experiment, which involves online control. In online control synthesis, the optimization time horizon denotes the duration within which control decisions are formulated, representing the period during which the system state is observed. Generally, this horizon is defined as a specific H relative to the current moment.

Our system includes uncontrollable environmental components (e.g., the rain component), so we are extending our offline validated model based on HMDP to online synthesis. As a substitute for learning the complete strategy σ^H_{opt} with significant computational resources, we utilized Uppaal Stratego periodically to conduct online strategy synthesis for the stormwater detention ponds but now using an optimization time horizon $h < H$.

For online control, we use the U.S. Environmental Protection Agency Stormwater Management Model (EPA-SWMM) [8] to simulate the real world. EPA-SWMM is an open-source physics-based dynamic rainfall-runoff model that has been used for decades in the urban stormwater management community. EPA-SWMM can precisely trace the dynamics of flow conditions (e.g. stream flow and stormwater detention pond water level). We use pySWMM [25] for the interfacing of EPA-SWMM with Python. Finally, we combine pySWMM with Uppaal Stratego using the STOMPC framework [7], which facilitates the setup of model-predictive control using Uppaal Stratego and an external simulator.

In the experiments of online control, we synthesized strategies considering an optimization time horizon of $h = 12$ hours, which results in the following query for Uppaal Stratego.

```
strategy S = minE (cost) [<=12*H] {} -> {sensor_w, t}: <> (t==12*H)
```

Note that H in the Uppaal Stratego model represents 1 hour, not to confuse with H, which represents the synthesis horizon for offline control. At each step in the MPC scheme, a weather forecast was generated for the optimization time horizon h using historical data.

5.3.1. Qualitative results

Fig. 13 shows the simulation results of online controller synthesis. The left side shows the results of all learned strategies, ranging from having no noise on the sensor readings to the sensor noise with a standard deviation of 100. The right side shows the results of several static controllers (and the no-noise learned controller as a reference). While the online control simulations have run for the full period of September 5 to 23, 2019, Fig. 13 only shows the 7-day interval during which heavy rain was falling. This shows how each strategy deals with the heavy rain as well as the couple of days after it.

As can be observed from the left side of Fig. 13, all learned strategies are able to prevent overflow of the pond (the capacity of the pond is $W = 200$ cm), regardless of the level of sensor noise. This result is consistent with the results from offline control from the previously described events, so that effect carries over to the online control scenario. The results also show that there is a clear distinction after the heavy rain between sensors without noise and those with noise. The model without any sensor noise keeps the

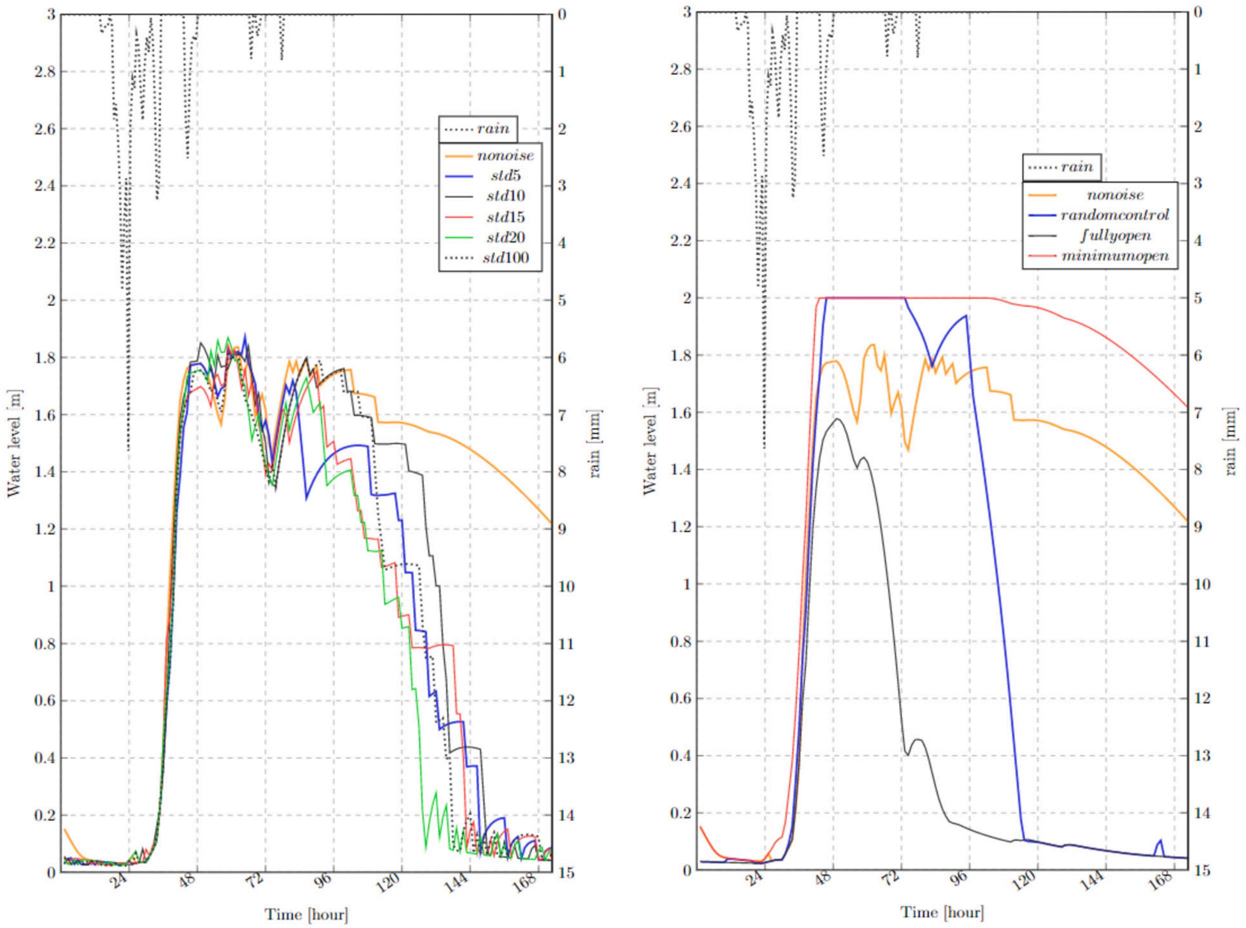

Fig. 13. Online synthesis results with 12 h optimization time horizon. The left figure shows the result of applying the learned strategies and for comparison, the right figure shows the results using static and random controllers.

stormwater longer in the detention pond, hence optimizing the pollutant sedimentation. All noisy sensors empty the pond quicker than the sensor without noise. Unfortunately, there is no clear distinction between the different levels of noise. This could mean that the learned strategies are somehow robust against noise on the sensor readings.

The right side of Fig. 13 effectively shows the extremes within which any learned strategy lies. A controller that always opens the orifice fully has a maximum water level that is approximately 0.4 m below the capacity water level W. On the other hand, a controller that always opens the orifice minimally results in an uninterrupted overflow duration of more than 2 days, which is definitely undesirable and unacceptable.

Model-predictive control is often praised for its robustness against model uncertainties [24]. The results in Fig. 13 demonstrate the robustness as well. One can see that the heaviest rainfall occurs around the $t = 24$ h, while there is a significant delay of several hours before the stormwater is actually entering the pond. While the differential equation (2) induces some delay between the rainfall and the inflow to the pond, the modeled delay is much less than the simulated delay by EPA-SWMM. Even though the delays do not match, repeatedly relearning a strategy creates robustness against modeling imprecisions.

In the next experiment, we changed the control horizon to $h = 6$. All other parameters of the system and the experiment remained unchanged. Fig. 14 shows the results for the case of no noise on the sensor readings (left) and for the case of a noisy sensor with standard deviation of 5 (right). The results of the sensors with higher noise levels are similar to the ones with the noise level of $\sigma = 5$. First, in both cases, the learned controllers are still able to prevent overflow. This indicates that a control horizon of $h = 6$ h is sufficiently large enough to observe the incoming rain on time. Observing the start of the raise of the water level more closely, one might see that the curves with the shorter control horizon start to go up earlier and then drop down before matching the curve for $h = 12$ h. The drop happens precisely at the moment a potential overflow could happen within the short horizon of $h = 6$ h. From that point on, reinforcement learning starts to adjust the orifice settings to let more water out of the stormwater detention pond.

The second observation is that the learned strategy for the case with no noise remains surprisingly almost unchanged. So shortening the control horizon did not affect the learned strategy. Having a shorter control horizon in practice is beneficial: as the calculation of each new strategy requires less computational effort with shorter control horizons, synthesizing a controller could then actually be implemented in embedded systems with limited hardware.

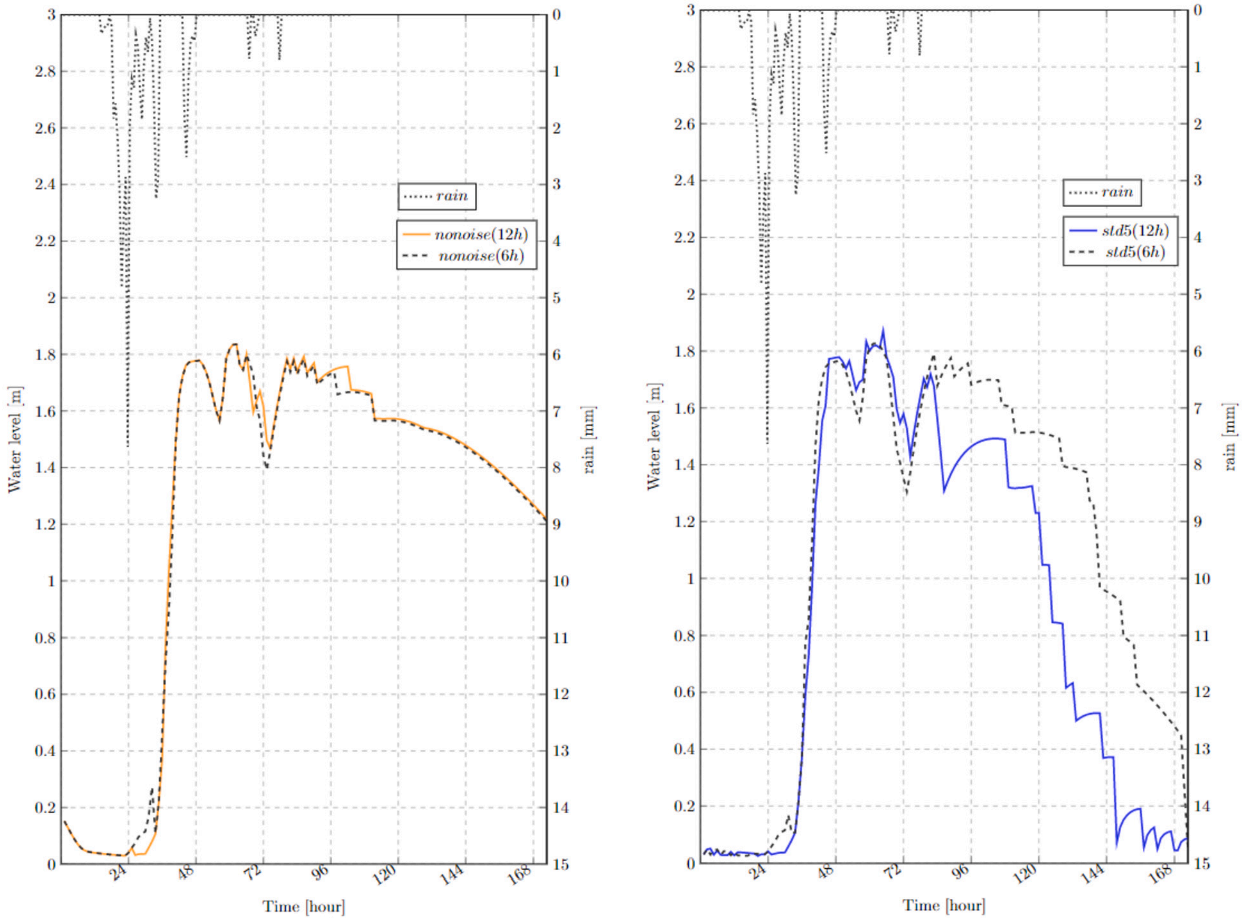

Fig. 14. Online synthesis results with 12 h and 6 h horizon under (left) no noise included observation (right) partial observation with noise ($\sigma = 5$).

Finally, shortening the control horizon did significantly effect strategies learned for sensors with noise (see the right plot in Fig. 14). The learned strategies keep the water lever higher once the heavy rain has passed by and no potential overflow is possible. The reason why a shorter control horizon gets more optimal results could be explained by the fact that reinforcement learning is used. Having a shorter control horizon means that shorter traces are generated from the model that feed the reinforcement learning algorithm. And shorter traces mean less variability in the traces is possible. Consider the number of possible control strategies. For a 6 h control horizon, there are 3^6 possible control strategies, significantly less than 3^{12} possible control strategies for a control horizon of 12 h. This effect is subsequently amplified by the stochastic weather forecast *and* the noisy sensors in the model. So, having less variability in the possible traces increases the probability that reinforcement learning learns the (near) optimal strategy. This could be the effect we are observing for the noisy sensors.

5.4. Control synthesis using memory-equipped sensors

In the following two sections we describe experiments covering two ideas on how to potentially improve the performance of controllers equipped with unreliable sensors. The first idea, described in this section, is about adding memory to the controller, while the second idea, described in Section 5.5, concerns using multiple sensors instead of a single one.

Sensors capable of retaining recorded data or information pertaining to a specific period of time can be highly advantageous, not only in situations where real-time data transmission is not feasible or practical, but also as an approximation of the entire state of the partially observed system [10].

In this section, we considered sensors equipped with a limited amount of memory for storing previously recorded sensor values. For the analysis, we rely on the SensorMemory model (see Section 4.5.1) that reads data every 2 minutes and stores the last 10 sensor readings, making them available for subsequent control strategy synthesis. The additional memory of the sensor will serve as a more fine-grained approximation of the entire state of the partially observed system, thereby providing a basis for improving the strategies being synthesized.

Table 4
Cost analysis based on synthesized control strategy under different levels of partial observation with memory.

	Cost [-]
No noise	731.143 ± 0.039
Gaussian Noise	
$\mu = 0$, $\sigma = 5.0$	732.056 ± 0.044
$\mu = 0$, $\sigma = 10.0$	732.191 ± 0.086
$\mu = 0$, $\sigma = 15.0$	734.747 ± 0.099
$\mu = 0$, $\sigma = 20.0$	734.832 ± 0.105
$\mu = 0$, $\sigma = 100.0$	736.506 ± 4.651

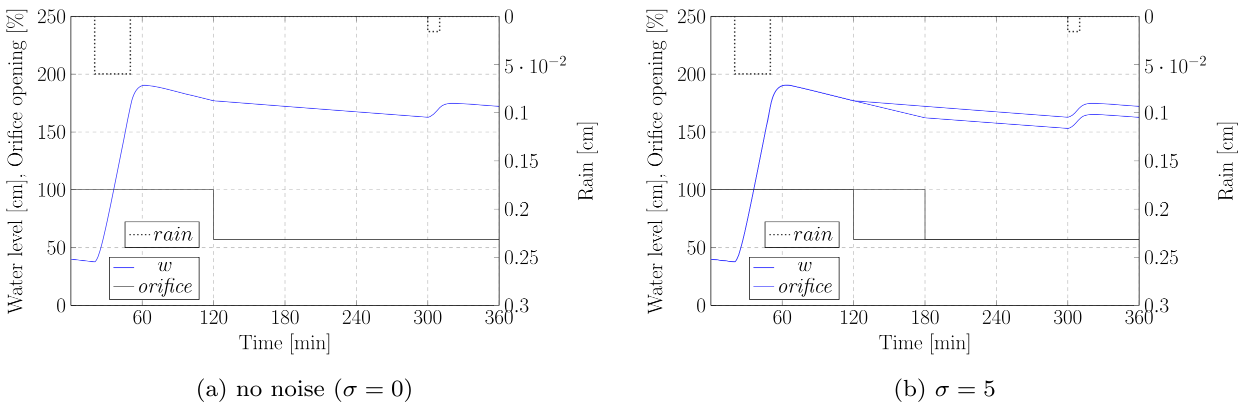

Fig. 15. Simulation results for a noiseless sensor with sensor memory (left) and a noisy sensor ($\sigma = 5$) with sensor memory (right).

5.4.1. Quantitative results

Table 4 shows the results for the synthesized strategies using sensors with the same noise levels, configurations and baselines as previously employed, except now have included memory into the sensor model. At first glance, the results seems reasonable, as the expected cost of each synthesized strategy grows monotonically with the noise level. But the intended effect, i.e., an improvement on the synthesized control strategy, is absent. When one compares the costs for the different strategies with those reported in Table 2, barely any improvement can be observed. For example, for the sensor with noise $\sigma = 10$, the cost value decreases from 732.55 ± 0.06 to 732.191 ± 0.086. While this is statistically significant using the confidence intervals, the improvement is nonexistent from a practical point of view.

5.4.2. Qualitative results

The results of applying a memory window of size ten to a noiseless observation during a synthesis process are depicted in Fig. 15. (While the figure only shows the result for a noiseless sensor and a noisy sensor with $\sigma = 5$, the conclusion also holds for the experiments with higher noise levels.) By comparing the controller's decisions in Fig. 15a to those in Fig. 10a, we can see that both controllers make very similar decisions. Specifically, Fig. 15(a) with memory applied does not fully open the orifice size at $300 \le t \le 360$, thereby allowing for slightly more sedimentation.

The synthesized strategy under the observation condition with noise for $\sigma = 5$ is depicted in Fig. 15(b). Compared to the observation in Fig. 15(a) without noise, inconsistent controller action can be observed at $120 \le t \le 180$, which shows that the ability to detect a decrease in water level appears weaker. On the other hand, comparing the results to Fig. 10b, a controller with memory seems to prefer a high orifice setting followed by a medium orifice setting, while a controller without memory prefers a high orifice setting followed by a low setting followed by a high setting. While the synthesized controllers are different, the reported costs in Tables 2 and 4 suggest that they have very similar accumulated sedimentation costs.

This leads us to conclude that adding sensor memory does not have a significant (and expected) improvement on the performance of the synthesized controllers.

5.5. Synthesis using observations of multiple sensors

As the detection accuracy of a sensor increases, the cost of the sensor also tends to increase. This poses a challenge in various industrial scenarios where a decision must be made between procuring a limited number of high-precision yet costly sensors or a larger quantity of low-cost yet imprecise ones. Consequently, sensor optimization becomes a critical aspect to consider.

Table 5
The cost comparison of the optimal strategy synthesis with observation under independently operated sensors and a single sensor.

sensor $\sigma = 20.0$	**Cost [-]**	**Overflow [min]**
single sensor	737.210 ± 0.040	≈ 0
two sensors	733.322 ± 0.047	≈ 0
four sensors	732.611 ± 0.139	≈ 0
six sensors	732.258 ± 0.129	≈ 0
eight sensors	731.057 ± 0.055	≈ 0

In this regard, there have been studies on methods to enhance sensor performance by utilizing multiple sensors [26]. Utilizing this method makes it possible to fill in missing data points. Multiple sensors can be used to compensate for each other's errors or limitations to improve the overall accuracy of observation [27].

We synthesized optimal strategies under different configurations of sensors. Our hypothesis is that synthesis with multiple independent sensors will produce better performance than synthesis of a single sensor observation. We implemented the experiment of this hypothesis using Gaussian noise $\sigma = 20$, and the result is shown in Table 5. The results show that 1) multiple sensors (all $\sigma = 20$) show better performance than that of single sensor with the same noise level and 2) each time adding an additional sensor increases the performance, but the effect diminishes with the number of sensors. The results demonstrate that the expected cost of the synthesized strategy using four low-quality sensors can result in a cost that closely approximates that of a single sensor with $\sigma = 10$, while using six sensors can result in a cost that closely approximates that of a single sensor with $\sigma = 5$. With these results, we can conclude that our hypothesis is correct.

6. Conclusion and future challenges

We introduce a method for synthesizing near-optimal control strategies for stormwater detention ponds, where the system state is only partially observed through noisy water level sensors. The synthesized control strategies adjusts the orifice size of the stormwater detention pond to control the outflow volume. We model the system as a hybrid Markov decision process and synthesize control strategies based on reinforcement learning as implemented in the tool UPPAAL STRATEGO. We conducted several experiments for analyzing the effect of generating control strategies under different levels of partial observability. One experiment confirms that including noise in the observations during synthesis increases the optimization cost of the synthesized strategy. On the other hand, online control results demonstrate that noisy sensors only affect the performance minimally. To cope with noisy observations, we performed experiments with memory equipped sensors as well as settings involving multiple sensors. While having a sensor with memory did not improve the synthesis performance, deploying multiple low quality sensors can achieve similar results as a single high-quality sensor.

For future research, we intend to develop safety-guaranteed and near-optimal strategies with the tool UPPAAL STRATEGO. By employing the state space partitioning method outlined in [28] or the abstraction method from [29], we could construct a guaranteed safe shield that ensures the satisfaction of the safety property φ. This involves calculating a collection of safe regions alongside the most permissive safety strategy σ_{φ}. Building on this foundation, our objective could then be to synthesize an optimal strategy σ_{opt} within a shielded state space in our current case study of a stormwater detention pond.

CRediT authorship contribution statement

Esther H. Kim: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation. **Martijn A. Goorden:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization. **Kim G. Larsen:** Writing – review & editing, Supervision, Conceptualization. **Thomas D. Nielsen:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Esther Hahyeon Kim reports financial support was provided by Digital Research Centre Denmark (DIREC). Martijn Angelo Goorden reports financial support was provided by Digital Research Centre Denmark (DIREC).

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT-3.5 to rephrase selected parts of the text. After using this tool/service, the authors reviewed and edited the content as needed and took full responsibility for the publication's content.

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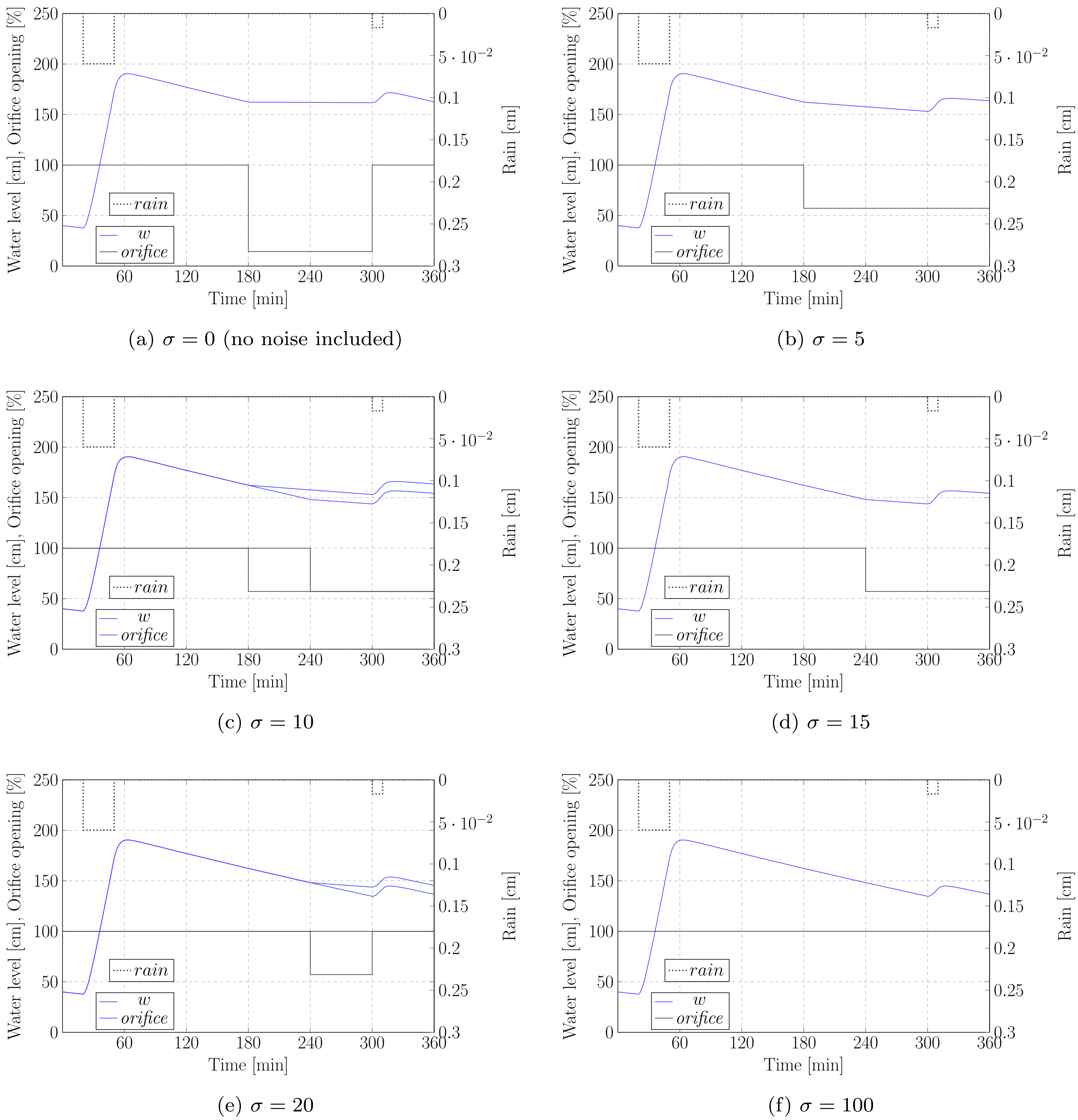
Appendix A. Additional experimental results

This appendix contains additional experimental results for the learning of offline controllers. Tables A.6 and A.7 show the new weather scenario, where the difference with the one in Section 5.2 is highlighted in bold font.

Table A.8 shows the new quantitative results, while Figs. A.16, A.17, and A.18 show the simulation plots, learned decision trees, and learned state-space partition's, respectively.



(a) $\sigma = 0$ (no noise included)

(b) $\sigma = 5$

(c) $\sigma = 10$

(d) $\sigma = 15$

(e) $\sigma = 20$

(f) $\sigma = 100$

Fig. A.16. Behavior of water (w) and orifice (opening rate (%) based on the synthesized controller with different levels of noisy sensors (each repeated 10 times).

Table A.6
The weather forecast used for learning in the additional experiments. The difference compared to the original weather scenario in Table 1 is highlighted in bold font.

State	Lower bound [min]	Upper bound [min]	Rain intensity [cm/min]
Dry	15	25	0
Raining	25	35	$0.05973 \pm \epsilon$
Dry	230	270	0
Raining	5	15	$\mathbf{0.01678} \pm \epsilon$
Dry	∞	∞	0

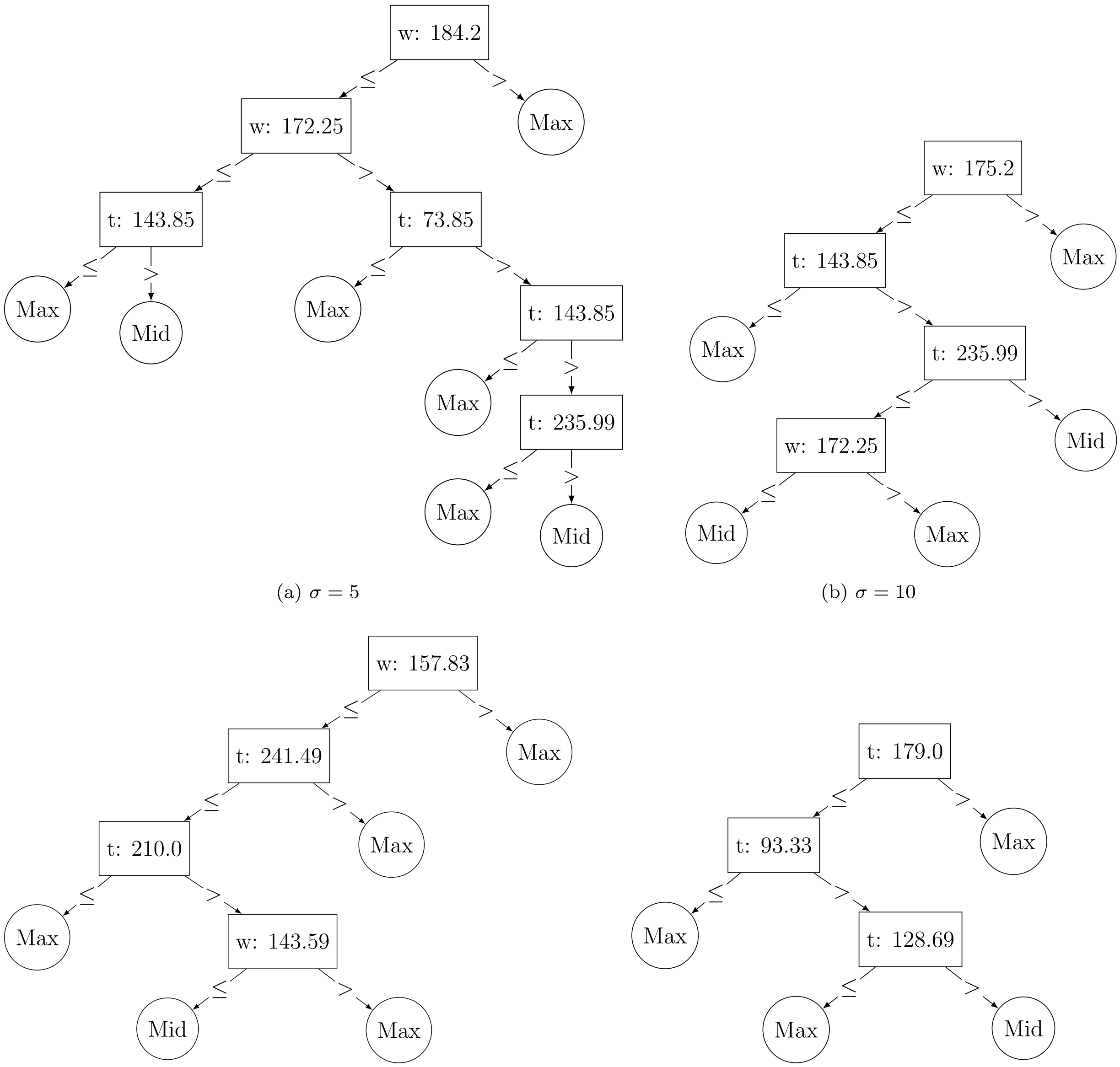


Fig. A.17. Decision tree representations of the synthesized strategies based on a high-quality sensor (a: $\sigma = 5$), two medium quality sensors (b: $\sigma = 10$) and (c: $\sigma = 20$), and an uninformative sensor (d: $\sigma = 100$).

Table A.7
The weather scenario utilized for evaluation. The difference compared to the original weather scenario in Table 3 is highlighted in bold font.

State	Lower bound [min]	Upper bound [min]	Rain intensity [cm/min]
Dry	20	20	0
Raining	30	30	0.05973
Dry	250	250	0
Raining	10	10	**0.01678**
Dry	50	50	0

Table A.8
New experimental results that outline the cost and duration of overflow for a range of learned control strategies for various degrees of sensor noise. For comparison, the table also includes results for a randomized controller and three static controllers.

Sensor Noise	Cost [-]	Overflow [min]
No noise	732.38 ± 0.04	≈ 0
$\sigma = 5.0$	735.70 ± 0.04	≈ 0
$\sigma = 10.0$	739.40 ± 0.11	≈ 0
$\sigma = 15.0$	742.64 ± 0.04	≈ 0
$\sigma = 20.0$	746.43 ± 0.13	≈ 0
$\sigma = 100.0$	748.03 ± 0.04	≈ 0
Random	23,581.8 ± 2087.26	1.86 ± 0.51
Static-max	748.03 ± 0.04	≈ 0
Static-med	12,136.4 ± 0.24	≈ 0
Static-min	202,427.0 ± 53.45	53.35 ± 0.05

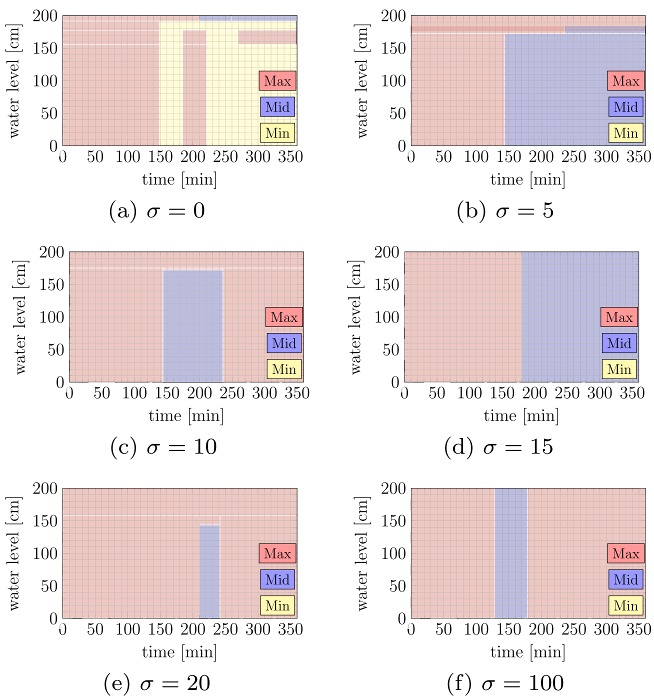


Fig. A.18. State-based partitioning of the synthesized strategies based on a no noise included (a: $\sigma = 0$), high quality sensor (b: $\sigma = 5$), three medium quality sensors (c: $\sigma = 10$), (d: $\sigma = 15$), (e: $\sigma = 20$), and an uninformative sensor (f: $\sigma = 100$).

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